

Mind Palace - Musical System Map

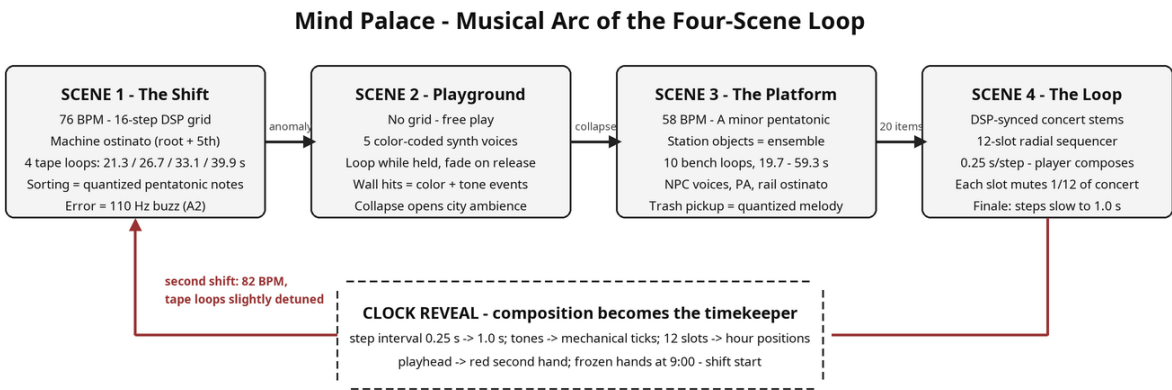
Generative, interactive VR composition for MUS 442 Senior Project II

Scene 1: The Shift

Scene	Time structure	Pitch / material	Player mapping
1. The Shift	76 BPM, 16-step DSP grid	Machine slices, A minor pentatonic	Sorting adds quantized notes
2. Playground	No global grid	Five color-coded synth voices	Holding and throwing cubes sounds color
3. Platform	58 BPM plus incommensurate periods	Station ensemble in A minor pentatonic	Compulsory cleanup adds melody
4. Concert / Loop	DSP-synced stems; 12-step sequencer	Six timbres, A3-A5 pitch reservoir	Slots replace concert, then clock

1. Musical Arc of the Four-Scene Loop

The full experience moves from externally measured work to free play, then to social routine, then to a player-authored loop that becomes the factory clock.



All scenes share one pitch language (minor pentatonic rooted on A) and sample-accurate AudioSettings.dspTime scheduling.

2. Scene 1: Factory Grid and Tape Loops

The factory score combines a sample-accurate 16-step ostinato with long tape-loop layers sliced from machine recordings.

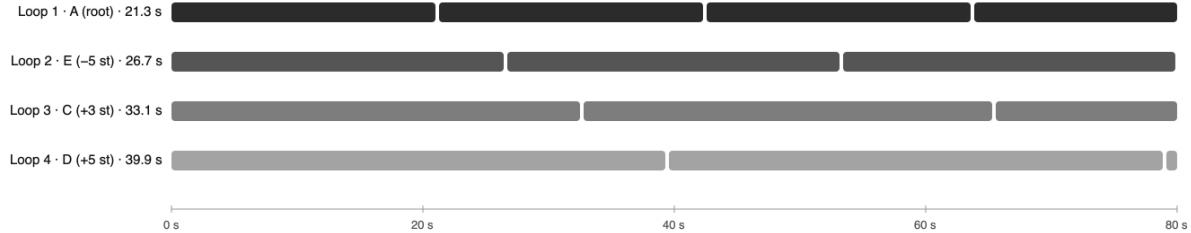
Scene 1 — Factory Score: Ostinato Grid and Incommensurate Tape Loops

Machine ostinato — one bar of 16 sixteenth-steps at 76 BPM (step = 0.197 s)



Percussive transients sliced at runtime from factory machine recordings; root and fifth alternate, octave accent closes the bar.

Four tape loops — periods never align (shown over 80 seconds)



Each loop is anchored to a different machine object in the room. Like Music for Airports, the four periods are incommensurate: the texture never repeats exactly.

3. Scene 3: The Station as Ensemble

The metro distributes musical responsibility across architecture, NPCs, PA speakers, trains, trash objects, and compulsory cleanup actions.

Scene 3 — The Station as Ensemble (58 BPM, A minor pentatonic)

Layer	Sound source	Time structure	Pitch content
1 · Rail ostinato	Track objects	16-step grid, sparse hits	Deep metallic A, low register
2 · Bench pads	10 Metalbench objects	Periods 19.7 → 59.3 s, free-running	Pentatonic dyads (table below)
3 · Lamp drone	Fluorescent lighting fixtures	Continuous hum	A2 + E3 + A3 (root, 5th, octave)
4 · Trash-can ticks	Each MetalTrashCan	Sparse 2-step pattern from index	Pentatonic, low register
5 · NPC voices	Each commuter body	Periods 11.3 / 13.7 / 17.9 / 19.3 / 23.7 / 29.3 s	Hum, murmur, whisper
6 · Tunnel rumble	Tunnel mouths	Continuous, distance-attenuated	Unpitched low noise
7 · PA announcements	Platform speakers	Triggered by train events + idle timer	TTS speech, ducked mix
8 · Trash melody	Player action (cart deposit)	Quantized to next 16th step	Rising pentatonic line, 2 octaves

Bench dyads (semitones above A3 → pitches):

{0, 7} = A3+E4 {3, 12} = C4+A4 {5, 12} = D4+A4 {7, 15} = E4+C5 {10, 19} = G4+E5

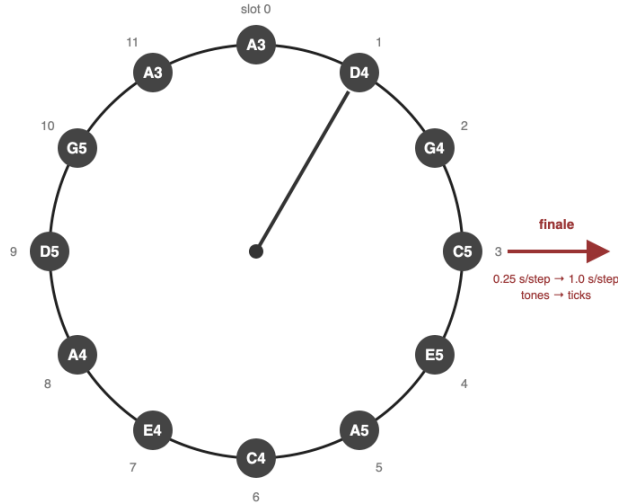
Every layer is scheduled on the same dspTime grid but repeats at its own incommensurate period — coherent, never identical, like the commuter cycle itself.

4. Scene 4: Sequencer as Clock Face

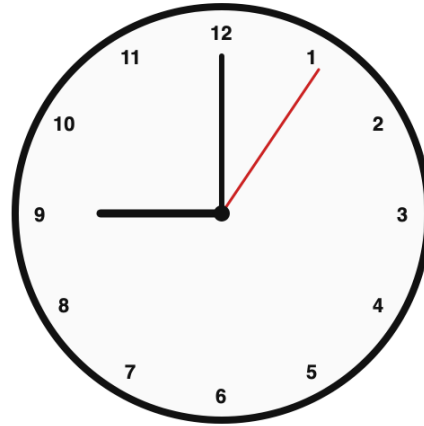
The finale uses the same circular interface twice: first as a musical sequencer and then as the literal clock that returns the player to the factory.

Scene 4 — The Radial Sequencer is a Clock Face in Disguise

Sequencer mode — step 0.25 s, six synth timbres



Clock mode — one tick per second, hands frozen at 9:00

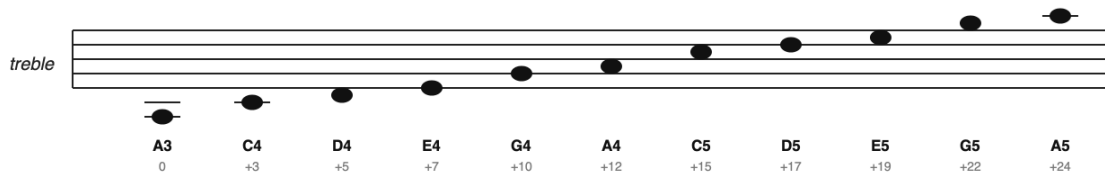


5. Shared Pitch Reservoir

A minor pentatonic functions as a common musical reservoir across the factory melody, station ensemble, and sequencer orbs.

Shared Pitch Reservoir — A Minor Pentatonic, A3 to A5

Semitones above A3 (220 Hz): 0 · 3 · 5 · 7 · 10 · 12 · 15 · 17 · 19 · 22 · 24 — used by the factory melody, metro ensemble, and sequencer orbs



Scene 1 tape loops transpose the same set below the root: A (0), E (-5), C (+3), D (+5). The error buzz is A2 (110 Hz); the lamp drone stacks A2 + E3 + A3.

Notation and Authorship Note

This document functions as notation for the implemented system. Conventional staff notation shows pitch material, but the composition itself is encoded in runtime scheduling, object mappings, scene transitions, and player-triggered rules. The project-specific interaction design, sound behavior, scene logic, generative music systems, and narrative integration were authored for Mind Palace by Semiha Paksoy.